Design Brief – Review 2

0. Header

Project code	FA26SE002 (group GFA26SE32)
Project name	Drone Delivery and Smart Locker Management System (DDSLMS) – product name Lock.R
Team	Lê Thị Yến Vi (leader) · Trương Nguyễn Thái Bình · Nguyễn Quốc Bảo Huy · Trần Kim Nhã
Supervisor	Đặng Ngọc Minh Đức (DucDNM) · Đỗ Hữu Tiến (TienDH)
Repository link (tag review-2)	https://github.com/LockR-Tech (backend, frontend, mobile, iot, docs) – tag review-2
Version	1.0
Date	09/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
3.2 b–c, f–g; 3.3 Research	Battery-aware assignment, strategy comparison, research outputs	Removed; idle drone above minimum battery; Software & Hardware Products	Iteration 1: one real drone (BM03 No. 1–3)
3.2 c) Module 5	AI summaries of operational reports	Removed	BM03 No. 4
3.2 d) NFR	p95 < 300 ms, availability > 95 %, telemetry 3–5 s	N1–N6 (§7)	BM03 No. 5
3.2 b), c) Module 2	Parcel collection by PIN/QR; generate PIN and QR codes	PIN at kiosk; app opens via Bluetooth or locker QR	BM03 No. 6
3.2 b–c, g; 3.3 b)	Modules 1–4 without a separate incident module	Added Module 3: Incident Reporting before Maintenance (Modules 4–5)	BM03 No. 7

1.2 Out of Scope

Item	Reason
Beyond-visual-line-of-sight commercial drone flights	Needs aviation permits; demo flights run in a controlled area with one drone.
Payment gateways other than SePay VietQR (MoMo, VNPay, cards)	One reconciled payment path; VietQR works with every Vietnamese bank app (BR-21).
Publishing on Google Play / App Store	Pilot ships as an Android APK and mobile web.
Battery-aware drone assignment research and strategy comparison	Removed from the registration (BM03 No. 1–3): one real drone gives no meaningful comparison.
Physical drone flight control	The drone Pi only reads autopilot telemetry and drives the release servo; it never arms or steers the drone.
More than one physical cabinet	One real cabinet (CAB-TU01, 7 boxes); other lockers run the same controller software simulated.

2. System at a Glance

Lock.R lets people send and store parcels without meeting in person. A customer rents a locker box or books a drone delivery in the mobile app and pays by VietQR; for a drone delivery the nearest technician on shift collects the parcel at

home and the drone drops it into the receiver locker's drone box. Boxes open with a 6-digit PIN at the kiosk, or from the phone over Bluetooth or by scanning the locker QR. Users are customers and receivers (mobile app, mobile web, kiosk), locker and drone technicians handling incidents and maintenance (mobile app), and administrators assigning work and following revenue (admin web). Each locker is a Raspberry Pi driving relays, door sensors and the drone-box lid over MQTT; Spring Boot microservices sit behind an API gateway, with an AI assistant. Core value: 24/7 contact-free hand-over with every box state and payment traceable. Architecture: SDD Figure 1.1.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Backend structure	Monolith; microservices	Microservices: 10 Spring Boot services, one database each (Flyway), gateway with JWT and roles, RabbitMQ events; members work per domain and failures stay local.	Nguyễn Quốc Bảo Huy	SDD Fig. 1.1
D2	Locker control and box opening	Arduino/RS485; vendor controller; Pi on GPIO	Pi 5 on GPIO with a versioned MQTT contract (ACL per device, resend outbox). Orders change only on the cabinet's report; Bluetooth tokens (Ed25519, 120 s, single use) are checked offline on the Pi.	Nguyễn Quốc Bảo Huy	UC16, UC34; BR-09, BR-24
D3	Payment method	MoMo / VNPay; wallet; SePay VietQR	SePay VietQR only: PAID only when the webhook amount covers the total; duplicate webhooks ignored; no merchant contract, works with every bank app.	Trương Nguyễn Thái Bình	UC08; BR-02, BR-21
D4	Drone delivery orchestration	Auto nearest-drone dispatch; technician accepts offer	The nearest free technician on shift collects the parcel at home, picks an idle drone, completes the loading checklist and launches; one delivery state machine fed by MQTT telemetry; a person answers for every flight.	Lê Thị Yến Vi	UC10, UC31; BR-10, BR-11
D5	Incident and maintenance tickets	Complaint to admin; box FAULT at once; ticket with SLA	Reports, failed inspections and drone faults become tickets; a customer report sets the box PENDING_CHECK, not FAULT; only a technician on shift claims (SLA 4 h, max 2 extensions); the scheduler escalates.	Trần Kim Nhã	UC12, UC27; BR-18, BR-23

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Cabinet reports a wrong door state (sensor misread, door stays shut)	Box shown empty while holding a parcel; false JAMMED	20 opens per box on CAB-TU01 with sensor log; hourly box/order reconciliation; emergency open	Nguyễn Quốc Bảo Huy
R2	Real drone flight cannot be shown reliably (weather, battery, permits)	Drone workflow not shown end to end	Weather check blocks bookings; without flight the drone Pi reads the locker ArUco marker, the lid opens, the servo releases and the lid closes with the same events; recorded flight as backup	Lê Thị Yến Vi
R3	External services throttle or delay (Firebase SMS, SePay webhook)	Sign-in fails or orders stay UNPAID in the demo	Firebase test numbers, email OTP fallback; payment polling; service check script before demos	Trương Nguyễn Thái Bình

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	Seven tables and ten columns added since 29/10 missing from Logical ERD (SR-10, SR-11)	rp3/srs-assets/drawio/05*	Regenerate Logical ERD	Trương Nguyễn Thái Bình	15/11/2026

ID	Gap	Location	Plan	Owner	Due
G2	Opening an order box via Bluetooth (fallback: locker QR or PIN) is built, not in SRS	SRS UC15; SDD §3; backend #137	Add use case, rule, sequence diagram	Nguyễn Quốc Bảo Huy	15/11/2026
G3	State machines and activity diagrams not yet in Report 3/4; 06c lacks pickup	docs/diagrams/src, srs-assets/drawio	Insert 5 + 5 diagrams; redraw 06c	Lê Thị Yến Vi	15/11/2026
G4	Report 4 declares no environments; Azure VM (api.locker-drone.tech) is staging	Report 4; backend scripts/seed-*.sql	Declare VM as staging with seed/rollback	Nguyễn Quốc Bảo Huy	15/11/2026
G5	Report 5 written; CI tests not mapped to §6 TC IDs and iot has no CI (BM02 SR-20)	Report 5; test suites	Map CI tests to §6 TC IDs; add iot CI	Trần Kim Nhã	22/11/2026
G6	SRS FRs have no testable acceptance criteria (BM02 SR-08; SR-13, SR-17 = G3; SR-01 fixed)	SRS §3.2–3.6	Add acceptance criteria per FR from §6	Trần Kim Nhã	22/11/2026
G7	Inspection activity 06g covers lockers only; drone schedules are built	srs-assets/drawio/06g	Add the drone path to 06g	Trương Nguyễn Thái Bình	15/11/2026

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Sign in, admin 2FA	User, Admin	Wrong password: MSG06, no token; admin needs a 6-digit email OTP valid 300 s.	TC-F01-01..03	Done
F-02	Create send order	Customer	Only AVAILABLE boxes of an online locker; box RESERVED; unpaid after 15 min: cancelled, box freed.	TC-F02-01..03	Done
F-03	VietQR payment	Customer	PAID only when received >= total; a duplicate webhook counts once; partial shows the remainder.	TC-F03-01..03	Done
F-04	Drop parcel, share code	Customer	Box opens only if PAID; drop gives STORING / OCCUPIED and a 6-digit PIN to the receiver.	TC-F04-01..03	Done
F-05	Pick up at kiosk	Receiver	Right PIN opens and completes the order; 5 wrong PINs lock the box 15 min (MSG21).	TC-F05-01..03	Done
F-06	Rent box by hours	Customer	1–720 h; multi-use PIN until the end; extension <= 24 h; ending frees the box.	TC-F06-01..03	Done
F-07	Open box by Bluetooth	Customer, Receiver	Owner/receiver token only, 120 s, single use; state changes on Pi report; fallback: QR or PIN.	TC-F07-01..03	Done
F-08	Overdue fee, expiry	Customer, Scheduler	500 VND/h, max 50,000 VND and 50 % of price; opens only after paying; > 72 h: EXPIRED.	TC-F08-01..03	Done
F-09	Cabinet commands	Locker Controller	No ack in 20 s: TIMEOUT, never success; relay ran but door closed: JAMMED; events resent.	TC-F09-01..03	Done
F-10	Book drone delivery	Customer	Refused: no free DRONE box (MSG13), > 5,000 g, > 5,000 m, unsafe weather or > 3 open orders.	TC-F10-01..03	Done
F-11	Assign drone, load, launch	Drone Technician	Only an idle drone > 20 % battery is assigned; launch needs the checklist; + 50 g re-prices.	TC-F11-01..03	Done
F-12	Track drone, drop into locker	Customer, Receiver	Stage updates in 2 s; on arrival lid opens, servo releases, lid closes; STORING with PIN.	TC-F12-01..03	Done
F-13	Report box incident	Customer	Title and description required, <= 5 photos; box PENDING_CHECK, no new orders, fee stops.	TC-F13-01..03	Done

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-14	Ticket within SLA	Locker Tech., Admin	Claim only on shift; SLA 4 h, max 2 extensions with reason; 3 overdue tickets block claims.	TC-F14-01..03	Done
F-15	Parcel drop and recovery	Drone/Locker Tech., Admin	Camera evidence required; recovery SLA 4 h; 60 % compensation needs admin and customer approval.	TC-F15-01..03	In progress
F-16	AI assistant	User	Answers cite sources; order questions see only own orders; reply within 10 s.	TC-F16-01..03	Done
F-17	Home pickup by technician	Drone Tech., Admin	Nearest free technician on shift; 180 s timeout: next one; pickup needs sender code within 300 m.	TC-F17-01..03	Done
F-18	Maintenance schedule	Admin	One locker or drone; matching technician role; drone schedule sets MAINTENANCE until inspected.	TC-F18-01..03	Done
F-19	Inspection checklist	Locker / Drone Tech.	Assigned technician only; items Pass/Fail/NA; all pass: next due date; fail opens ticket, FAULT.	TC-F19-01..03	Done
F-20	Dashboard, revenue report	Admin	Revenue = payments collected minus refunds in the period; each total opens the entries behind it.	TC-F20-01..03	Done

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
N1	Performance	p95 API response time	Read <= 1 s, write <= 2 s at 100 sessions	k6 on staging, 10 min
N2	Responsiveness	PIN / "Open box" to door-open result	<= 5 s in 95 % of 50 opens	Timed run on CAB-TU01 + logs
N3	Realtime	Order status / drone stage delay to app	p95 <= 2 s	Event vs receive time, 30 events
N4	Payment accuracy	Double-counted replayed webhooks	0 of 100	Replay test in CI
N5	Security	PIN lockout; Bluetooth token reuse	Lock after 5 wrong PINs; 100 % reused tokens rejected	Automated tests in CI
N6	Availability	Monthly API uptime	>= 99 %	Health check every 5 min

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Lê Thị Yến Vi	Leader; drone delivery	1) Drone booking, dispatch, home pickup; 2) Drone telemetry, drone Pi (ArUco, servo); 3) Dashboard, revenue, drone screens	WF-01	D4; order-service (drone), drone-iot
Trương Nguyễn Thái Bình	Mobile – customer, payment, incidents	1) Incident report and technician ticket screens; 2) SePay VietQR payment; 3) Send / rent / my-orders screens, rental rules	WF-03	D3; payment-service, mobile
Nguyễn Quốc Bảo Huy	Backend, IoT, DevOps	1) Architecture, gateway/auth, CI/CD, VM; 2) Locker controller, kiosk, MQTT, Bluetooth open; 3) AI assistant and knowledge base	WF-02	D1, D2; auth, iot, assistant services; iot repo
Trần Kim Nhã	Drone fleet, maintenance	1) Drone units, devices, logs, tickets; 2) Maintenance schedules, inspections, shifts; 3) Parcel drop, overdue and expiry	WF-04	D5; locker-service (drone), order-service

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Drone delivery with home pickup	Customer, Drone Technician, Receiver, Locker Controller	Customer books a drone delivery	Parcel dropped into the receiver locker and picked up	F-03, F-05, F-10–F-12, F-15, F-17, F-20	Order AWAITING_DISPATCH → STORING → COMPLETED; Stage AWAITING_DISPATCH → PICKUP_ASSIGNED → PICKED_UP → … → READY_FOR_PICKUP; Drone IDLE → RESERVED → IN_FLIGHT → IDLE	Lê Thị Yến Vi
WF-02	Rent a box or send a parcel via locker	Customer, Receiver, Locker Controller, System Scheduler	Customer rents a box or creates a send order	Items collected and box released, or EXPIRED after the overdue fee	F-01–F-09, F-16, F-20	Order INITIALIZED → STORING → COMPLETED / EXPIRED; Box AVAILABLE → RESERVED → OCCUPIED → AVAILABLE	Nguyễn Quốc Bảo Huy
WF-03	Box incident to resolution	Customer, Locker Technician, Admin	Customer reports a faulty box	Ticket RESOLVED with the box back in service or locked, or REJECTED with feedback	F-13, F-14	Box AVAILABLE / OCCUPIED → PENDING_CHECK → previous (rejected) or FAULT → previous (resolved); Ticket OPEN → IN_PROGRESS → RESOLVED / REJECTED	Trương Nguyễn Thái Bình
WF-04	Maintenance of lockers and drones	Admin, Locker / Drone Technician	Admin creates a maintenance schedule or a drone ticket and assigns a technician	Inspection recorded; asset back in service, or a repair ticket opened	F-14, F-18, F-19	Drone IDLE → MAINTENANCE → IDLE / FAULT → IDLE; Box AVAILABLE → FAULT → AVAILABLE; Ticket OPEN → IN_PROGRESS → RESOLVED	Trần Kim Nhã

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	WF-01	Main	Receiver locker CAB-TU01 has a free drone box; idle drone > 20 % battery; technician on shift; safe weather	Book with home pickup → pay → nearest technician accepts, collects at home (code, photo), loads, launches → ArUco read, lid opens, servo releases, lid closes → receiver enters PIN → admin opens revenue	Order STORING with PIN, then COMPLETED; log shows servo release; payment in revenue	Lê Thị Yến Vi
DS-01b	WF-01	Exception	Receiver locker with no free drone box; parcel of 6,000 g	Book a drone delivery	Booking refused (MSG13 / weight limit)	Lê Thị Yến Vi
DS-02a	WF-02	Main	CAB-TU01 has a free Standard box; customer near it with Bluetooth on	Choose a box, rent 2 h → pay by VietQR → PIN issued → open with PIN at kiosk, close → reopen with "Open box" (Bluetooth) → end rental → admin opens revenue	Order COMPLETED, box AVAILABLE; payment in revenue	Nguyễn Quốc Bảo Huy
DS-02b	WF-02	Exception	Rental DS-02a in STORING; a second rental past its end time	Enter a wrong PIN 5 times at the kiosk; open the overdue rental	Box locked 15 min (MSG21); overdue box opens only after the fee is paid	Nguyễn Quốc Bảo Huy
DS-03a	WF-03	Main	Occupied box on CAB-TU01; locker technician on shift	Customer reports with a photo → technician claims → inspects, fixes on site with a photo → resolves	Box PENDING_CHECK → back in service; ticket RESOLVED; customer notified	Trương Nguyễn Thái Bình
DS-03b	WF-03	Exception	Report on a box with no fault; technician X holds 3 overdue tickets	X claims it; a technician on shift claims, rejects without, then with feedback	Claim refused; reject needs feedback; ticket REJECTED, box back to its previous state	Trương Nguyễn Thái Bình
DS-04a	WF-04	Main	Drone DRONE-S550-07 idle, no mission; drone technician account	Admin creates a schedule with a checklist and assigns the technician → technician is notified, marks every item Pass with a photo	Drone MAINTENANCE (no deliveries) until the inspection, then IDLE; next due date set	Trần Kim Nhã
DS-04b	WF-04	Exception	Schedule for CAB-TU01 assigned to technician A	Technician B completes it; A marks one item Fail on a box	B refused (assigned technician only); ticket for A, box FAULT, due date kept	Trần Kim Nhã

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	20	3	60

Source	Count	Minimum per item	Subtotal
Main workflows (end-to-end)	4	1	4
State transitions of main entities	57	1	57
Agreed NFRs	6	1	6
Committed total			127

Automated in CI: at least 50% of the committed total.